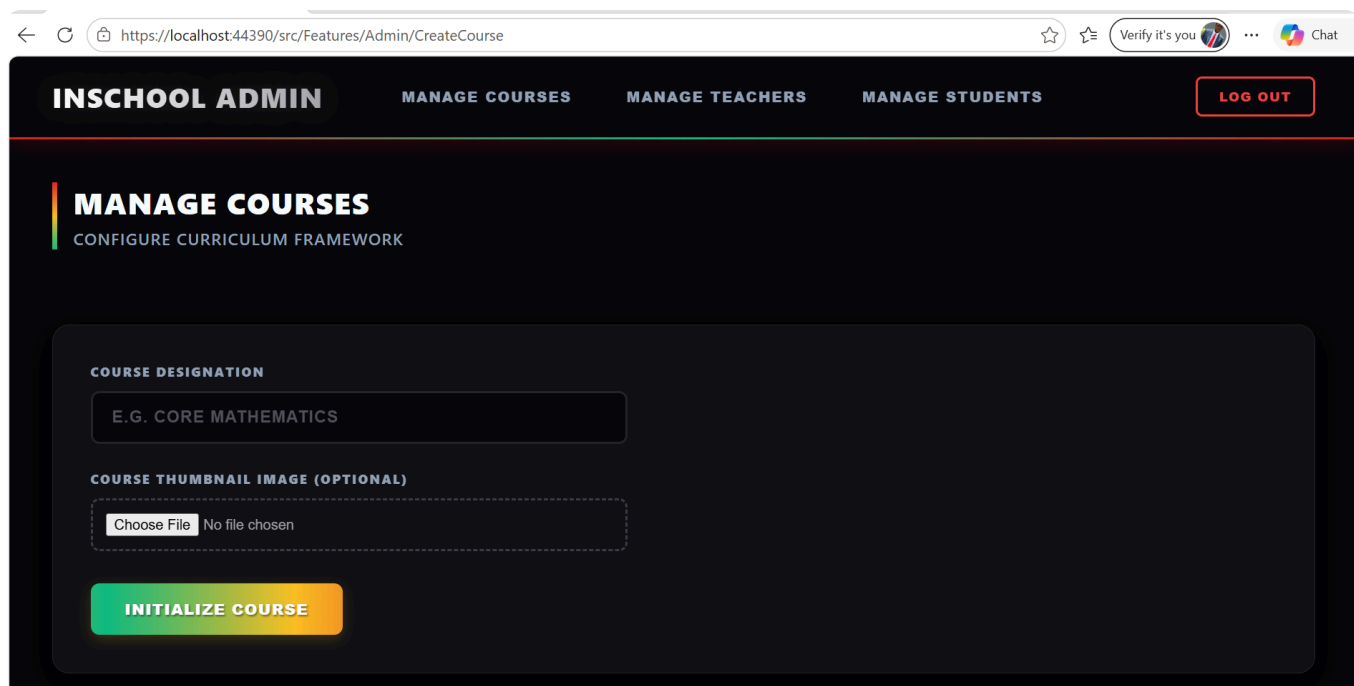


Admin Module - Simple Guide



The screenshot shows a web browser window with the URL `https://localhost:44390/src/Features/Admin/CreateCourse`. The page has a dark theme. At the top, there is a navigation bar with the text "INSCHOOL ADMIN" and three links: "MANAGE COURSES", "MANAGE TEACHERS", and "MANAGE STUDENTS". A "LOG OUT" button is located on the right. Below the navigation bar, the "MANAGE COURSES" section is active, with the subtitle "CONFIGURE CURRICULUM FRAMEWORK". The main content area contains a form with two sections: "COURSE DESIGNATION" with a text input field containing "E.G. CORE MATHEMATICS", and "COURSE THUMBNAIL IMAGE (OPTIONAL)" with a "Choose File" button and the text "No file chosen". At the bottom of the form is a green "INITIALIZE COURSE" button.

What Admins Do to Set Up the System

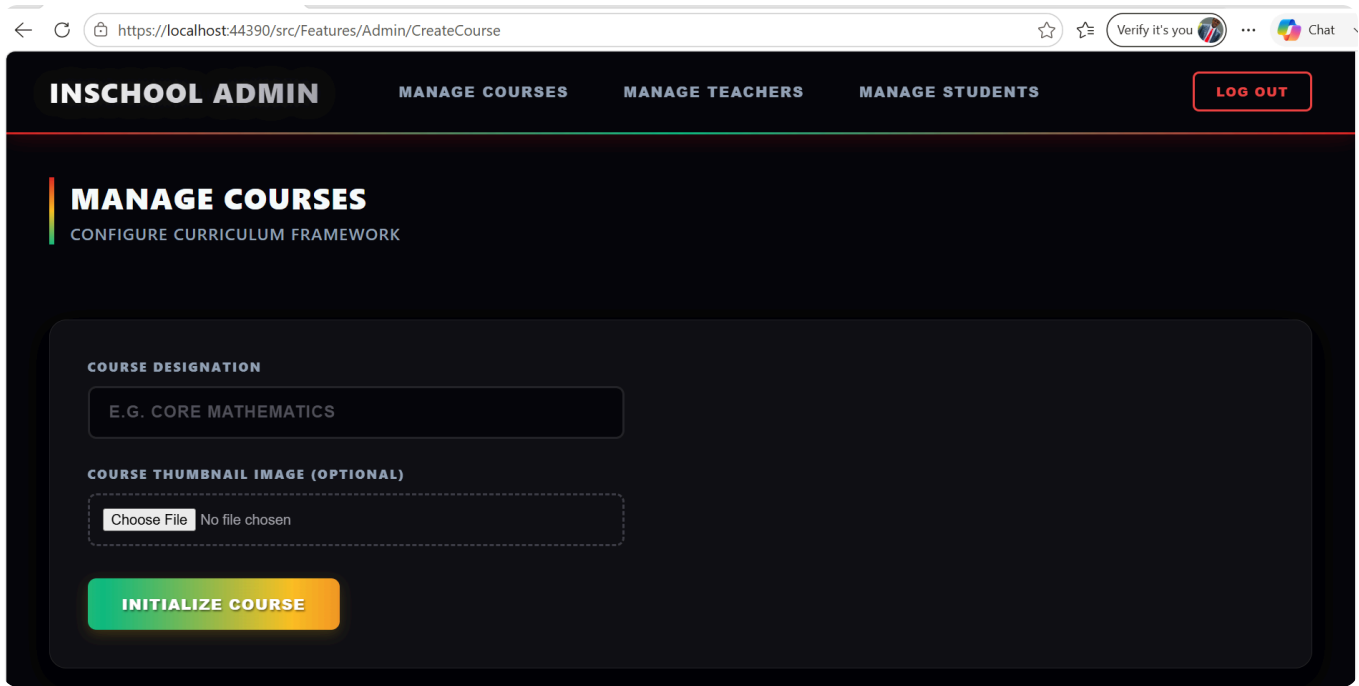
Introduction

The **Admin Module** is like the **control center** of the school system. Admins use it to:

- ☒ Add new courses
- ☒ Hire and assign teachers to courses
- ☒ Enroll students in bulk (all at once, not one-by-one)

This guide explains each of these tasks in simple terms.

Activity 1: Adding New Courses



What Happens

An administrator wants to add a new course called "Mathematics 101" to the system.

1. Admin goes to "Add Course" page
2. Types the course name: "Mathematics 101"
3. Uploads a course picture (thumbnail)
4. Clicks "Create Course"
5. The system saves everything to the database
6. The course appears in the student portal

Step-by-Step Explanation

Step 1: Enter Course Name

The admin types the course name in a text box.

Why we protect this:

- The system checks: "Did you actually type something?" (not blank)
- The system checks: "Is this course name already in the system?" (no duplicates)
- Only valid course names get saved

Step 2: Upload Course Picture

The admin uploads a picture that will show when students browse courses.

Why we protect this:

- **File type check**: Only images allowed (not Word docs or random files)
- **File size check**: Image can't be too big (max 5MB)
- **Unique file naming**: The system renames files automatically so two admins uploading "math-picture.png" don't overwrite each other
 - Example: Instead of "math-picture.png", the system saves it as "3f2504e0_20240115_math-picture.png"
 - Why? This prevents accidents and security problems

Step 3: Save to Database

The admin clicks "Create Course" and the system:

1. Checks that the course name is valid
2. Checks that the picture file is safe
3. Saves the course to the database
4. Saves the picture to the server

Step 4: See Your Work

A grid (table) shows all courses that have been created. The admin can:

- See the course name
- See the course picture

- See when it was created
- Edit or delete it



Why This Approach?

What We Do	Why It Matters
Check for blank names	Prevents incomplete data in the system
Check for duplicate names	No two courses with the same name
Check file type	Only images, no viruses or bad files
Rename files automatically	Prevents accidental overwrites
Show all courses in a table	Admin can instantly see what was created



Activity 2: Hiring Teachers & Assigning Them to Courses

← ↻ https://localhost:44390/src/Features/Admin/CreateTeacher 🔍 ☆ ⚙️ Verify it's you ... Chat

INSCHOOL ADMINMANAGE COURSESMANAGE TEACHERSMANAGE STUDENTSLOG OUT

MANAGE TEACHERS & ASSIGNMENTS
CREATE INSTRUCTOR PROFILES AND ASSIGN THEM TO ACTIVE COURSES.

1. INITIALIZE TEACHER RECORD

FULL NAME
E.G. MR. MENSAH

USERNAME
E.G. MENSAH_TEACH

INITIAL PASSWORD

CREATE PROFILE

2. ASSIGN TO COURSE MATRIX

SELECT TEACHER
-- Select Teacher --

SELECT COURSE
-- Select Course --

CONFIRM ASSIGNMENT

What Happens

An administrator wants to:

1. Hire a new teacher (create a teacher account)
2. Assign that teacher to specific courses



Part A: Creating a Teacher Account

The admin creates a new teacher in the system.

What the admin enters:

- **Username:** A unique login name (e.g., "jsmith")
- **Password:** A secure password (e.g., "SecurePass123!")
- **Full Name:** The teacher's real name (e.g., "John Smith")

What the system checks:

- ☒ All fields are filled in (no blanks)
- ☒ Username is unique (not already taken)
- ☒ Password is strong (at least 8 characters, uppercase and lowercase)
- ☒ Once verified, the password is encrypted (locked away, not readable)

Why?

If we didn't check these things, we could accidentally create two teachers with the same username, or a weak password that's easy to hack.



Part B: Assigning Teacher to Course

Now the admin assigns the teacher to specific courses.

The admin selects:

- **Teacher**: "John Smith" (from a dropdown list)
- **Course**: "Mathematics 101" (from another dropdown list)
- Click "Assign"

What happens behind the scenes:

The system records this relationship in the database:

John Smith → teaches → Mathematics 101

What the system prevents:

- **Duplicate assignments:** If John Smith is already assigned to Mathematics 101, the system stops the second assignment
 - Why? We don't want the same teacher-course link twice in the database

What shows up:

A table displays all teacher-course assignments:

Teacher	Course	Assigned Date
John Smith	Mathematics 101	Jan 15, 2024
Sarah Johnson	English 102	Jan 15, 2024

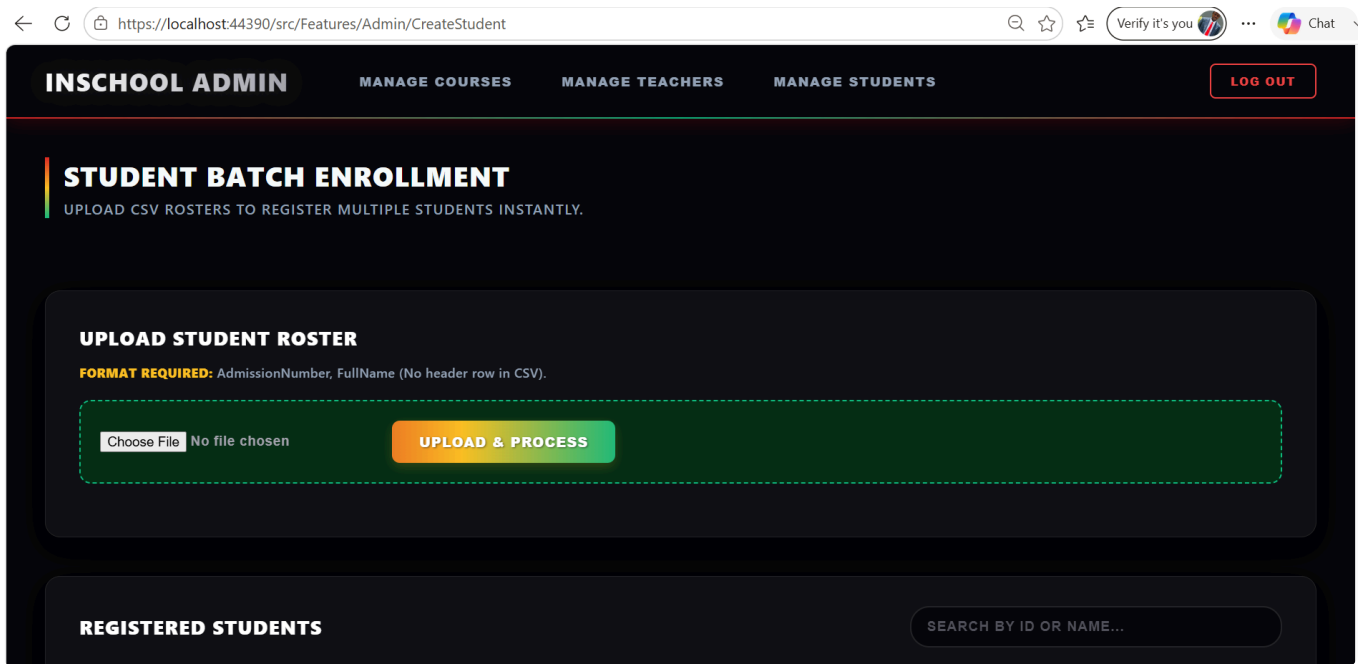


Why This Approach?

What We Do	Why It Matters
Check username is unique	No two teachers can have the same login
Require strong password	Teachers can't use weak passwords like "123456"
Encrypt passwords	Even database admins can't see passwords in plain text
Prevent duplicate assignments	Teacher isn't accidentally assigned to same course twice
Show all assignments in table	Admin can see who teaches what at a glance



Activity 3: Enrolling Many Students at Once



What Happens

Instead of typing each student one-by-one, the admin uploads a file with all students.

The Old Way (Painful):

Admin types: "Student 1, Student 2, Student 3..." × 500 students = **16+ hours of work**

The New Way (Fast):

Admin uploads one CSV file with 500 students = **Done in 1 minute**



What's a CSV File?

A CSV file is just a spreadsheet saved as plain text. It looks like this:

```
AdmissionNumber,FullName,Email
ADM001,Alice Johnson,alice@school.edu
ADM002,Bob Smith,bob@school.edu
ADM003,Catherine Lee,catherine@school.edu
ADM004,David Chen,david@school.edu
ADM005,Emma Rodriguez,emma@school.edu
```

The admin creates this file in Excel, then uploads it to the system.



Step-by-Step: Uploading Students

Step 1: Upload the File

Admin clicks "Upload CSV" and selects the student file.

Step 2: System Checks Each Line

The system reads each row and checks:

- ☒ Is the admission number blank? (Stop if yes)
- ☒ Is the student name blank? (Stop if yes)
- ☒ Is the email blank? (Stop if yes)
- ☒ Is this student already in the system? (Skip if yes)
- ☒ Is all the information correct?

Step 3: Create Student Accounts

For each valid student, the system:

1. Creates a login account (username = admission number)
2. Generates a temporary password (student changes it on first login)
3. Saves the student to the database

4. Shows success message

Example Result:

Successfully enrolled 485 students
(15 rows had errors or duplicates – see details below)



Part B: Finding a Student (Instant Search)

Once students are enrolled, the admin can search for them instantly.

How it works:

- Admin types in the search box: "Alice"
- The system instantly filters the table to show only "Alice Johnson"
- No waiting, no server delay, instant response

Why is this fast?

The search happens on the admin's computer, not on the server. It's like using the browser's Find function (Ctrl+F).



Part C: Deactivating a Student

If a student drops out, the admin can deactivate their account.

How it works:

1. Admin clicks "Deactivate" on a student's row
2. System shows a confirmation message: "Are you sure you want to deactivate this student?"
3. Admin confirms
4. Student account becomes INACTIVE (they can't log in)
5. Table updates to show "INACTIVE" status in red

Why the confirmation?

To prevent accidents. If admin accidentally clicks "Deactivate", they see a warning before it happens.



Student Status Display

Each student shows their current status:

Status	Color	Meaning
ACTIVE	Green	Student can log in
INACTIVE	Red	Student account is disabled



Why This Approach?

What We Do	Why It Matters
Accept CSV files	Saves 95% of admin time (500 students in 1 min, not 16 hours)

What We Do	Why It Matters
Check each row	Bad data is caught before it enters the system
Check for duplicates	Same student isn't enrolled twice by accident
Generate temporary passwords	Secure by default; student changes on first login
Instant search (no server wait)	Admin finds students in milliseconds
Confirmation modal	Prevents accidental deletion of student records
Color-coded status	Easy to see who's active vs. deactivated at a glance



Complete Admin Workflow

Admin logs in



- Create Course
 - Enter course name (checked for blanks/duplicates)
 - Upload thumbnail (checked for file type, size)
 - Save to database
 - GridView shows all courses
- Create Teacher
 - Enter username, password, name
 - System validates uniqueness and strength
 - Password is encrypted
 - Account created
 - Assign teacher to course(s)
 - System checks for duplicate assignments
 - GridView shows all teacher-course mappings
- Enroll Students
 - Upload CSV file
 - System checks each row
 - System prevents duplicates
 - Creates accounts for all valid students
 - GridView shows all students
 - Admin searches instantly (no server wait)
 - Admin can deactivate students with confirmation

Key Features Explained Simply

Bulk Upload = Time Saver

Instead of clicking "Add Student" 500 times, the admin uploads one file. Done.

Instant Search = No Waiting

When searching for a student, the result appears instantly on their screen. No waiting for the server.

Automatic Checks = Prevents Mistakes

- System prevents duplicate courses
- System prevents duplicate teacher assignments
- System prevents duplicate student enrollments
- System confirms before deleting anything

Secure = Built In

- Passwords are encrypted
- Bad files are blocked
- Important operations ask for confirmation
- Duplicate data is prevented at the database level

What the Admin Portal Does

The Admin Module is the **setup wizard** for the entire school system. It handles:

1. **Course Setup**: Create all courses with pictures
2. **Teacher Setup**: Create teacher accounts, assign them to courses
3. **Student Setup**: Bulk enroll students, manage their accounts

Without the Admin Module, the system would have:

-  No courses
-  No teachers
-  No students

With the Admin Module, the system has:

-  Organized courses with pictures
-  Teachers assigned to the right classes
-  Hundreds of students enrolled in minutes



When Things Go Wrong (Error Messages)

The system tells admins clearly when something is wrong:

Situation	What Admin Sees
Blank course name	"Course name cannot be empty."
Duplicate course name	"A course with this name already exists."
Bad file type (e.g., .docx instead of image)	"Only image files (.jpg, .png, .gif) are allowed."
File too large	"File size cannot exceed 5MB."
Blank username	"Username cannot be empty."
Weak password	"Password must be at least 8 characters with uppercase and lowercase."
Duplicate teacher assignment	"This teacher is already assigned to this course."
Bad CSV row	"Line 15: Missing required fields."
Duplicate student enrollment	"Student ADM001 already enrolled."

Situation	What Admin Sees
Accidental delete attempt	Confirmation modal: "Are you sure?"



The Presenter's Simple Pitch

When presenting the Admin Module, say:



// "The Admin Module is the **control center** that sets up the entire school system.

For Courses: Admins add new courses with pictures. The system checks that course names aren't blank or duplicated, and that pictures are real images (not viruses or random files).

For Teachers: Admins create teacher accounts with secure passwords. The system automatically assigns teachers to specific courses and prevents the same teacher from being assigned to the same course twice by accident.

For Students: Instead of typing hundreds of students one-by-one (which would take days), admins upload one CSV file and all students are enrolled in minutes. The system checks each student record and prevents duplicates. Admins can also search for students instantly and deactivate accounts if needed.

The key is that **the system does the hard work**: validating data, preventing errors, confirming important actions, and letting admins see their work immediately in organized tables."



Summary Table

Task	What Admin Does	What System Does	Result
Add Course	Types name + uploads picture	Validates, checks duplicates, renames file	Course visible to students
Create Teacher	Enters name, username, password	Encrypts password, checks uniqueness	Teacher can log in
Assign Teacher	Selects teacher and course	Prevents duplicates, links in database	Teacher sees assigned courses
Enroll Students	Uploads CSV file	Reads file, checks each row, creates accounts	500 students in 1 minute
Search Students	Types in search box	Instant filtering on their screen	Find student in seconds
Deactivate Student	Clicks deactivate, confirms	Updates database	Student can't log in



Why This Design Works

- ✓ **Fast:** Bulk operations instead of manual clicking
- ✓ **Safe:** Automatic checking prevents errors
- ✓ **Clear:** Organized tables show what's been created
- ✓ **Secure:** Passwords encrypted, bad files blocked
- ✓ **Easy:** Simple confirmation modals prevent accidents
- ✓ **Instant:** Search results appear without waiting

The Admin Module turns what would be days of work into hours, and what would be error-prone manual entry into automated, validated processes.